STELLAR CYBER MODULAR SENSOR INSTALL AND CONFIG ON ESXI TRANSCRIPT

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L01 - INTRODUCTION

L01A-PRES: Introduction (03m:40s)

00:00-01:00

Hi and welcome to the *Stellar Cyber Open XDR Modular Sensor Installation and Configuration* module. I'm David Shirley from Stellar Cyber's Enablement and Training teams. The lessons in this module show you how to install, configure, and verify a Stellar Cyber Modular Sensor in VMware's ESXi. Hypervisor. We will cover log forwarding and network traffic features of the Modular Sensor. Throughout this module there may be concepts and terms you may not be familiar with, but that's okay and to be expected. The important thing is that you become familiar with the concepts and terms described. Through the Stellar Cyber Open XDR training curriculum, your understanding and application of cybersecurity principles will deepen with each repetition, iteration, and example.

01:00-02:00

As you advance, you're on your way to becoming a cybersecurity superstar. Thank you for your dedication. Welcome to *Stellar Cyber Open XDR Modular Sensor Installation and Configuration*. The following lessons are included in this module:

- *Overview*: At a high level we will see where the Modular Sensor fits into the Stellar Cyber Open XDR platform. We'll also take a high-level look at the available features, the sizing requirements, and the firewall communication requirements.
- *VMware ESXi Configuration:* This lesson starts with an overview of the ESXi existing networking configuration and the creation of a new ESXi port group prior to the creation and the editing of the Modular Sensor virtual machine.

02:00-03:00

- *Modular Sensor Command-Line Interface Configuration:* With the Modular Sensor virtual machine installed, next we will configure the Modular Sensor from the command-line interface, including changing the initial login and entering and validating the networking host name and configuration manager settings.
- *Modular Sensor DP configuration:* With the configuration of the Modular Sensor complete, the next step is authorizing the Modular Sensor in the Data Processor's user interface.
- *Modular Sensor Configuration Validation*: In this lesson, we validate the Modular Sensor and Data Processor configuration by looking at the syslog traffic arriving to both the Modular Sensor and the Data Processor.
- *Modular Sensor Traffic-Monitoring Configuration*: This lesson adds the network traffic feature to ingest network SPAN traffic. The required ESXi configuration was completed in an earlier lesson.

03:00-03:40

To complete the configuration, we will edit the sensor profile that the Modular Sensor is bound to.

- *Modular Sensor Traffic Monitoring Validation:* This lesson focuses on validating the configuration of the network traffic feature by looking at both the Modular Sensor CLI and the Data Processor user interface for increases in counters and seeing the records in the **Investigate | Threat Hunting** page.

Now let's get going and happy learning!

L02 - OVERVIEW

L02A-PRES: Overview (23m:55s)

00:00-01:00
The first lesson in the *Stellar Cyber Open XDR Modular Sensor Installation and Configuration* module is *Overview. Overview:* In this lesson, we'll take a high-level view to see where the Modular Sensor fits into the Stellar Cyber Open XDR platform. We'll also take a look at the available features, the sizing requirements, and the firewall-communication requirements.

A Modular Sensor lets you easily add the feature you like to your sensor, which helps simplify your deployment. You can choose which features for which Modular Sensors you have enabled on different Modular Sensors, enables you to manage the VM–the virtual-machine requirements–from a hardware perspective as in from a resourcing perspective, for the Modular Sensors again based on the features that they use,

01:00–02:00
so increasing the CPUs or the memory or the disk space required based on the features you have enabled. A Modular Sensor is also recommended by Stellar Cyber when you want to deploy an Aggregator. Simply enable the Aggregator feature in the Modular Sensor profile. It simplifies your network architecture in the deployment.

Overview of Features: The Modular Sensor features are a **Log Forwarder**, which are on by default. **Network Traffic** to look at the actual traffic passing through the sensor we often hear that called SPAN traffic. **Sandboxing** Of the traffic that's passing past the sensor, the SPAN traffic, if we can reassemble packets and look at files that are attached then we can scan those. **IDS** part of the sandboxing feature, an **Aggregator** to allow us to have a connector use the Modular Sensor to forward traffic back to the Data Processor.

02:00-03:00
That enables us to have fewer firewall ports open from different devices in your network back to the Data Processor. They aggregate via a Modular Sensor. The **Tenable Nessus**: We can configure that and we'll show in detail in another lesson on how to do that. Note: Modular Sensor features. They're configured not in the Modular Sensor, but in the profile that the sensor uses. We create a sensor profile, and then bind that to the actual Modular Sensor. **Log Forwarder**: The Log Forwarder feature enables Modular Sensor to receive log traffic from network and security devices and forward those to the Data Processor. As we said before, this feature is enabled by default and cannot be turned off. **Network Traffic**: This feature is enabled on the Modular Sensor to ingest local or virtual Network Traffic.

03:00-04:00
If we have our Modular Sensor virtualized in something like ESXi, as we've been doing, we can see that it can monitor traffic within the virtual switch or if we have a hardware Modular Sensor, we can look at the traffic on the network. The Modular Sensor in the virtualized can also look at that physical Network Traffic if we have a NIC card bound to that virtual switch. **Common practice** is the dedicated interface on the Modular Sensor for traffic monitoring is connected to a SPAN port–a switched port analyzer–a port that can carry multiple VLANs or carry mirrored traffic from multiple ports back to that virtual switch or back to the port on our Modular Sensor. Here are our sizing requirements. From a virtualized perspective, if we've got this sensor in ESXi or Hyper V or some other environment, there is a table in the Knowledge Base where we can get all of this information.

04:00-05:00
We can see here, log forwards 500 events per second, one virtual core, three gig of RAM, and 64 gig of SSD. You see if we have an Aggregator it's the same thing for 500 events per second or a hundred meg of traffic. Stellar Cyber recommends that you provision at least one and a half times as much memory as the number of CPU cores to ensure stable performance. It's very important to adhere to the sizing requirements to make sure that these sensors perform as expected. The Modular Sensor virtual machine must meet the minimum specifications in these tables. Let's go through a couple more of these tables. We can see here that **CPU** in the first column, **RAM** in the second column, and disk space. Log Collector and Log Forwarder you can see that is on for all of these rows. That is the same for any Modular Sensor.

05:00-06:00
We can't turn log forwarding off. Then Network Traffic, you may have a Modular Sensor that is doing log collection, but not doing Network Traffic and not doing IDS, not doing sandbox, not doing Tenable. You can see there one CPU, three gig of RAM, and 23 gig of

disc. Now as you increase or turn features on in the Modular Sensor profile that the Modular Sensor is bound to, sorry I shouldn't have said the Modular Sensor profile. It's just a sensor profile for Modular Sensors. You can see when we start to turn those on in the second row it needs additional CPU requirements because we now have Network Traffic turned on. As you go down that table and increase the number of features that are turned on for that Modular Sensor, you can see that the requirement for the CPU and the RAM and the disk space increases. Make sure you have sufficient resources available for the features that you have turned on.

06:00-07:00
Modular Sensors are not just virtualized sensors in ESXi or Hyper V. We have a line of physical devices called the photon: the Photon 160, the Photon 250 and the Photon 400. You can see here by looking at this table of which one for which use. You need to work out how much Network Traffic throughput you're going to be looking at and then the number of interfaces required you can see. Photon 160 you can just see there as an example, up to 500 meg of Network Traffic, and it has six RJ45 1-gig ports on it. If your requirement is for more than that, then you need to step up to the 250 or even the 400. You can see Network Traffic capture throughput of the network interface requirements for a location are used to determine the Modular Sensor model to deploy. Again, this table is available in the documentation.

07:00-08:00
Please look in the Knowledge Base to make sure that you choose the right physical photon sensor for the environment and the design of your network. Now we have some terminology to go through, which will help you along the way in your journey of understanding Stellar Cyber and how it all works.

Telemetry is collected by Modular Sensors to create Interflow records. Telemetry refers to the data collected about the behavior and actions with a network or a system. This includes detailed records of events such as system logs, data collected via API connections from security identity and infrastructure systems, and user activity logs. Telemetry data is rich and comprehensive, providing a real-time or near-real-time stream of information that can be analyzed to detect anomalies, track system performance, and identify potential security threats or breaches.

08:00-09:00
Collecting telemetry data allows security analysts to conduct in-depth investigations and respond to incidents with a high degree of precision.

As opposed to telemetry, we have metadata. Metadata is created by Modular Sensors when creating Interflow records. Metadata is about data; it is data about data. Metadata provides contextual information about other data being analyzed or stored and includes information about files such as creation dates, file sizes, file types, Network Traffic sources, and destination IP addresses, port numbers, protocols used, bytes transmitted and received, and TCP connection status. User activities including login times, resource access logs. While metadata does not contain the content of the data itself, collecting metadata is invaluable for security analysis

09:00-010:00
because it can help in identifying patterns, trends, and anomalies without the need to process the entire content of the data streams.

Looking at the two together, telemetry and metadata are crucial for cybersecurity because they serve different roles. Telemetry provides a granular view of events and behaviors within systems and networks, enabling detailed analysis and response. Where metadata offers contextual insights about the data, which can add in analysis, organization, and sometimes preserving privacy while still monitoring for security threats.

An Interflow record created by Stellar Cyber consists of telemetry and metadata from security, identity, and infrastructure systems.

Some more requirements regarding Modular Sensors. Here we're going to look at the requirements of firewalls.

10:00–11:00
This is: What do I need to open between the Modular Sensor and the Data Processor or the Modular Sensor and something else? We can see in the first column–the **Source** column–we only have Modular Sensor. Again, this table is available in the Knowledge Base in the documentation. What we're looking at here is a source of a Modular Sensor and a destination. The destination we'll see as we go through the different lines in this table will change, but for a Modular Sensor to communicate with a Data Lake, it needs to have port 443 open. This of course is if you have a firewall sitting between your Modular Sensor and your Data Lake. It needs to have 443 open with protocol TCP. Those are required, absolutely. For the Modular Sensor to get to the Data Lake it also needs to have ports 6640 to 6648 open with TCP with TLS version 1.2. That's for communicating to the configuration manager.

11:00-12:00
A Modular Sensor to a Data Lake 8443. That's TCP or HTTP with TLS 1.2. That's for it to download software and files from the Data Processor. Confirm that your firewall policies

allow traffic from this Modular Sensor to the destination with a port and protocol specified in these tables. More information so Modular Sensor to the DA, or the Data Analyzer. I won't read through all these, but we can see it's there: port 8888 and 8889 is required. A Modular Sensor, if it has an Aggregator enabled, needs to also have port 8080 open and then to get to the archive.Ubuntu.com, it needs to have port 443 open. That's for software updates. Again, confirm that those are open if you have a firewall between your Modular Sensor and your Data Processor or your Data Analyzer or your Ubuntu.

12:00-13:00

Just a couple more requirements to have open. If you have your firewall fairly closed, these need to be open. Confirm that that is all correct in your firewall rule settings. A couple more. Modular Sensors collect Network Traffic traversing a network via SPAN ports on Ethernet switches. The Network Traffic collected includes north/south traffic. This is network communication to and from clients and servers accessing the internet. In our diagram, this is the red dotted line from the laptop in our bottom left of the network diagram up through the switch connected to another switch through the firewall, out through the router out to the internet. This is termed north/south traffic. We then have east/west traffic or lateral traffic. This is network communication among internal clients and servers.

13:00-14:00

Again, our laptop on the bottom left is communicating through its switch to a core switch and then back out to another access switch where the servers are connected. This is east/west traffic. We can see that our Modular Sensor is connected to the core switch via two ports: 1/1 on the switch and 1/2. One/one on the core switch is connected to our Ethernet0, our management interface, on our Modular Sensor and Ethernet1. On the Modular Sensor the SPAN port is connected to port 1/2 on the network switch. SPAN ports, or switch port analyzer ports, are configured on switches to replicate or mirror Network Traffic from specific switch ports or VLANs connected to the SPAN port. We can mirror either one or more ports on the switch out to the SPAN port or we can mirror one or more VLANs from the switch out to that port.

14:00-15:00

Stellar Cyber Modular Sensors receive that traffic from the switch SPAN port for analysis. They perform deep packet inspection (DPI) to identify applications and run IDS signatures to detect threats and look for malware. They forward network telemetry metadata as Interflow to the Stellar Cyber platform. We can see here with our red dotted line, this is our north/south traffic. It is being replicated out of that SPAN port, switch port 1/2 into our Ethernet1 on our Modular Sensor, our SPAN port. That's the north/south traffic. Now we have east/west traffic as well from our laptop to a server and we can see that that is also

being replicated to our Modular Sensor. Out of the switch port 1/2 into our same SPAN port, Ethernet1 on the Modular Sensor.

15:00-16:00

Let's look at a couple of features highlighting what's available for Modular Sensors via the features of a sensor profile. The first one is log forwarder - log filtering. Modular Sensors collect, pass, and normalize logs sent from security and infrastructure tools into Interflow records in JSON format and forward to the Stellar Cyber platform. Modular Sensors have hundreds of built-in parsers to interpret and normalize logs into Interflow records. Logs that are not relevant for the threat investigation can be filtered at the Modular Sensor to optimize the amount of data that is ingested on the Stellar Cyber platform. Another feature of the Stellar Cyber Modular Sensor is Network Traffic. The Modular Sensor captures Network Traffic and creates metadata summaries

16:00-17:00

that are normalized into Interflow records in JSON format and forwarded to the Stellar Cyber platform. Modular Sensors identify Network Traffic flows and applications in Network Traffic using deep-packet inspection (DPI) that is normalized into Interflow records in JSON format and forwarded to the Stellar Cyber platform. IDS Modular Sensors apply IDS rules to network packets to identify threats and create metadata summaries that are normalized into Interflow records in JSON format, and forwarded to the Stellar Cyber platform. Malware. Stellar Cyber Malware Sandbox can be configured to identify malware in unencrypted files and create metadata summaries that are normalized into Interflow records in JSON format and forwarded to the Stellar Cyber platform.

17:00-18:00

What we mean by unencrypted format or unencrypted files is if there's traffic whether it's north/south or east/west, traffic that is in our switch, and it is unencrypted, the Stellar Cyber Modular Sensor is able to reassemble the files from the traffic that is going past or going past the networking SPAN interface and put them together to run signatures against them to see if those files are malicious. An Aggregator is another feature of the Stellar Cyber Modular Sensor. It can be configured as an Aggregator for the traffic from other Modular Sensors or other server sensors, Window sensors, and Linux sensors within an organization. It gives us the ability to configure a Modular Sensor with an Aggregator function. Communication from sensors with the Stellar Cyber platform is the Aggregator instead of each individual sensor.

18:00-19:00

What we mean by that is if you have multiple Modular Sensors in your remote network, and you have multiple server sensors, whether they're Windows sensors or Linux sensors, rather

than having all of those sensors communicate directly through the firewall to the Data Processor, we can aggregate that communication. We could have the feature turned on on one of the Modular Sensors. All of the other Modular Sensors in that remote network would communicate directly to our Modular Sensor with the Aggregator turned on. We can have all of our server sensors, Linux and Windows, communicate directly to the sensor with the Aggregator function turned on. Through the firewall, only one set of rules is enabled allowing the Modular Sensor with the Aggregator feature turned on to communicate with the Data Processor rather than all of them doing that individually. For Tenable, we can turn on and install and run Tenable.io Nessus scanner on our Modular Sensor.

19:00-20:00

With a license from Tenable.io, a security scan can be initiated to run on a Modular Sensor to scan the network for vulnerabilities. Along with performing vulnerability scans, a Tenable.io connector can be configured to collect telemetry from Tenable that is normalized into Interflow records. Connectors: Modular Sensors can obtain telemetry data from infrastructure identity and security systems via connectors using APIs, normalize those into Interflow records in JSON format and then forward them to the Stellar Cyber platform. Using connectors on Modular Sensors enables the ability to trigger response actions to infrastructure identity and security tools from the Stellar Cyber platform via automated or manual action.

20:00-21:00

The final feature to talk about with Modular Sensors is buffering. Modular Sensors can be configured to buffer Interflow records that are cleared after being sent to the Stellar Cyber platform. If there is a connection issue to the Stellar Cyber platform preventing the Modular Sensor from forwarding Interflow records, the Modular Sensors can temporarily hold the buffered Interflow record, and that until the connection is resumed to prevent loss of that data. In more detail let's talk about the log forward and the log filter. The log forward and log filter overview logs are sent to the Modular Sensor from security and network infrastructure systems. We can see there we've got logs coming into our Stellar Cyber Modular Sensor. They are normalized into Interflow records in JSON format and then forwarded to our Stellar Cyber platform.

21:00-22:00

The Stellar Cyber platform uses machine learning, statistical analysis rules, heuristic rules, custom ATH playbooks, and graph machine learning to identify security threats, generate Alerts, and correlate those Alerts into Cases. Log filters can be configured on Modular Sensors to optimize the data ingested by the Stellar Cyber platform by excluding specific log types that are not required for threat investigation or detection. Examples of security and infrastructure systems include firewalls, EDR, NDR, vulnerability management, email,

and many more configured to send logs in the form of syslog: two different formats CEF format and LEEF format, JSON format, Netflow and IPFix format to the Stellar Cyber Modular Sensor. We have all of these different devices on our network, forwarding log traffic to the Stellar Cyber Modular Sensor.

22:00-23:00
Modular Sensors normalize the logs from all these different sources into Interflow records in JSON format and forward those onto the Stellar Cyber platform. We can see there as the logs come in they are parsed, normalized, and sent in JSON format, as a normalized record to the Stellar Cyber platform. Note each log source can be parsed and normalized by Modular Sensors. That is found in the Stellar Cyber documentation and has a dedicated TCP/UDP port number for identification. Network Traffic analysis overview. Another feature will go into a little bit more detail on. Network Traffic analysis for network detection and response. Network Traffic from switches is sent to the Stellar Cyber Modular Sensor via a SPAN port or a TAP port.

23:00-23:55
A SPAN port is configured on a switch. A TAP device could be used if the switch does not support such a function. The Modular Sensors then use deep-packet inspection to identify applications, create a session state, correlate app processes, and add metadata such as the number of bytes used by the application during a session. A summary of the traffic and the metadata is normalized into Interflow records and forwarded to the Stellar Cyber platform.

The lesson *Overview* is complete. You should now be familiar with where the Modular Sensor fits into the Stellar Cyber Open XDR platform as well as the available features, the sizing requirements, and the firewall communication requirements of the Modular Sensor.

L02B-DEMO: Overview (05m:20s)

00:00-01:00
The first lesson in the *Stellar Cyber Modular Sensor Installation and Configuration* module is *Overview. Overview*, in this demonstration, we will look at three elements: Modular Sensor features, Modular Sensor sizing requirements, and Modular Sensor firewall requirements. First, we'll go into the sensor profile and look at the Modular Sensor features. Then we'll look in the Stellar Cyber documentation for the Modular Sensor sizing and firewall requirements. Let's get started.

For the Modular Sensor features we'll look at the Sensor profile. We'll go to **System | Sensor Profiles**, we'll create a Modular Sensor profile.

01:00-02:00
We can see here by default, the only one that's configured that we can't actually turn off is the **Log Forwarder** feature. The two we're going to be looking at in this lesson is the Log Forwarder and the Network Traffic. This is where you would turn Network Traffic on as well. Any of these you can click to expand and collapse. If we look at the Log Forwarder and expand it out the only one that's on by default is Buffering. We'll talk about these in a later lesson, and we will talk about Network Traffic in this lesson as well to turn that on for our Modular Sensor. We'd give it a name, choose a receiver, and click **Submit** but I just wanted to show you that in the Modular Sensor features, we turn these on and off in the profile, not the actual sensor. For the sizing requirements, we're going to look in the documentation. We're going to go to **Installing**. Expand that. **Best practices**.

02:00-03:00
Expand that. **Virtual Appliance Sizing** is what we'll select. Okay. Here we scroll down to **Modular Sensor Specifications**. We can see further down **Modular Sensor Specifications**. We can see here for a Log Forwarder if we have the Log Forwarder feature on for up to 500 events per second, it's just one virtual core, three gig of RAM, and 64 gig of SSD. If that starts to increase, if we do Log Forwarder and Aggregator, and we will talk about Aggregator in a different lesson, same requirements here. One virtual core, three gig of RAM, 64 gig of SSD. Okay. If that increases, and we saw that there's a possibility for that, the rule of thumb: Stellar Cyber recommends that you provision at least one and a half times as much memory as the number of CPU cores to ensure stable performance.

03:00-04:00
The example would be here when we're looking at the features, they all have the **Log Collector** feature. We can't turn that off. If we turn on **Network Traffic**–so our first example–if we just had Log Forwarder, and Network Traffic without IDS, without sandbox and without Tenable turned on we would need two CPUs, three gig of RAM, and 28 gig of disk. Here's a table that will show you some guidelines for the sizing requirements. For the firewall requirements, let's collapse this down. We're going to go down to **Reference**, expand **Reference**, **Firewall Requirements**. We're going to expand that. We're going to choose **General Firewall Requirements**. Okay. Remember we're talking about a Modular Sensor so the one we're interested in seeing is the code MS, which is talking about Modular Sensor.

04:00-05:00
If we look at all the ports that are required to be open, we scroll down looking for Modular Sensor. Here we see it:` Modular Sensor talking to the Data Lake requires port 6640 to 6648 and that protocol is TCP with TLS 1.2 to communicate with the configuration manager. You

have got to look at all of these line items where it has Modular Sensor, Modular Sensor, and if we've got it with an Aggregator, so a Modular Sensor talking to a Modular Sensor with an Aggregator turned on, that's what's required. These are your firewall requirements to turn on between having a Modular Sensor talking to all of these different elements, whether it's a Data Lake or the Data Analyzer or another Modular Sensor, with the Aggregator feature enabled.

05:00-05:20
The lesson overview is complete. You should now be familiar with Modular Sensor features, sizing requirements, and firewall requirements. Thank you.

L03 - ESXI CONFIGURATION

L03A-PRES: ESXi Configuration (09m:55s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Modular Sensor Installation and Configuration* module is *VMware ESXi Configuration. VMware ESXi Configuration*. In this lesson, we'll take a look at the existing ESXi networking configuration and discuss how to create a new ESXi port group.

For this module, the goal is to install a Modular Sensor in ESXi. This Modular Sensor will have multiple interfaces. We'll have a management interface for log forwarding and for management of course, and we'll have a network traffic interface. If we look in the diagram, our management and log forwarding interface, is the LAN interface or the blue interface coming out of the Modular Sensor and we're going to have a network traffic interface for network traffic analysis,

01:00-02:00
that is the orange interface or the SPAN interface coming out of our Modular Sensor. We're going to configure ESXi networking for this. We're going to have promiscuous mode to monitor all network traffic within that virtual switch. You can see there we have Windows and Linux servers we're going to need to see that traffic coming into our port on the Modular Sensor. Promiscuous mode is the way we configure that. Then we're going to validate that configuration. Both from a log forwarding perspective, we're going to have logs coming in from a firewall that is sitting in that network and from a network traffic perspective, we can see that we have Linux and Windows servers that will be chatting away and we want to see that traffic coming into whether it's north/south or east/west traffic

coming into that SPAN port on our Modular Sensor. The first thing to do in our ESXi environment is for the networking perspective.

02:00-03:00

Select **Networking** from the navigator panel on the left-hand side of the ESXi UI, and then from the **Port Groups** tab select **Add port group**, and we'll have this window pop up that we can see here. We need to enter a name for the port group. That is just a name. There's nothing special about that, something that means something to you in your network and in your environment. Enter a VLAN of 4095 for the VLAN ID. Now this is something in special to ESXi. 4095 is not a VLAN that's within the range of usable user VLANs. It's a special VLAN number in ESXi that we can go to in more detail later, but just the overview is, what this is saying is any device that is connecting to an interface on this port group and we've called this port group, “PortGroup-NDR-SPAN-10Net” is saying that it is capable of receiving tagged and untagged VLAN IDs.

03:00-04:00

What we mean by that is if it was a trunk port from a switch that was coming in, it would be tagged. Our 802.1Q frames would be tagged in this example, or there could be frames from that east/west traffic between two servers talking, which would be untagged frames. The 4095 is just saying that the device connected to this port group, which will be our Modular Sensor, is capable of receiving both tagged and untagged frames. There is our little note for that. Okay. What else do we need to do? We need to bind our port group, our PortGroup-NDR-SPAN-10Net, to a virtual switch. In this example, vSwitch-10Net is the example that this port group has been bound to. If we expand the **Security** section, this is the promiscuous mode that we have to set up, we can see there that we need to set our promiscuous mode to accept.

04:00-05:00

There is a different way of doing this, probably not recommended, but the reason there is accept and reject is you could take the point of view of we have a virtual switch here–vSwitch-10Net–that is acting as a switch. It is building a table of MAC addresses and port numbers and only communication between anything in that MAC table will be forwarded to a particular port. The alternative is that we could turn the switch to accept promiscuous mode, in which it would forward all frames to all interfaces. Then if you had a server, for example, that you didn't want to be receiving, and overwhelmed by all of those frames, you would choose **reject** for the interface that your servers use. We're going to do it the probably the “more normal” way, which is to have a switch behaving like a switch. We are configuring this port to be in promiscuous mode to accept all frames that are traversing that virtual switch, or vSwitch-10Net.

05:00-06:00

The final thing to do is to choose **Add** to add our new port group to the ESXi networking configuration. From a virtual machine perspective, we're going to go and create from the navigation panel on the left select **Create / Register Virtual Machine**, and this pop-up window will appear. This is the beginning of creating our Modular Sensor in a virtualized environment. From the pop-up menu, we've got to choose to deploy a virtual machine from an OVF or an OVA. We have an OVA file, which we'll be deploying. We choose that top option there, and then select **Next**. Enter virtual machine name. Again, this is just a name that means something to you within this ESXi deployment. We then need to select our software.

06:00-07:00

We're choosing to use 4.3.7 in this example, so that we can upgrade it to 5.1.1 in another lesson. Either you drag the file to that open space or if you click in that open space you're able to browse to where you have that aella-modular-ds-4.3.7 ova file previously saved. Select **Next**. Okay. It's now asking where would you like to store this virtual machine? Where will our Modular Sensor be stored? We're going to store it in our only store we have available here, so we select the Data Store and choose **Next**. For network mappings, we can see that we need to choose the network map for this. This is our interface for our management and our log forwarding interface. Network mapping is the part of the configuration. They're referring to the management interface of the Modular Sensor.

07:00-08:00

For the VM network, select the correct port group. We need to choose the right one. For **Disk provisioning** for our virtual machine we're going to choose **Thick**. We want to provision all of the disk space available at the machine creation. We don't want to choose **Thin** where it would create it as required. De-select the **Power on automatically** checkbox because we just might make some edits before we actually power on this virtual machine. When you've done that, choose **Next**. The last window you see is just the confirmation window to show you your summary of the virtual machine you're about to create. If you've made a mistake here, if the networking is incorrect click the **Back** button go make those changes, and when you're finished click **Finish** to continue. Your virtual machine will now be created. The first thing it's got to do is load from that OVA file all the files required to build your virtual machine.

08:00-09:00

Go down to the bottom and look at your recent tasks to monitor the progress of your virtual machine creation. Now depending on where that file is coming from, whether it's coming from a local resource or a remote resource, will just depend on how long it takes. But it can range from a few minutes to 10 or 15 minutes depending on your connection to the ESXi

host. When it's finished this is what we're left with. We're ready to power on a machine. But before powering on the machine, remember we unchecked **Power on** before it finished. Otherwise, it'll be booting right now. There are a couple of edits we just want to go in there and make sure of. Click **Edit**. Let's just go here and add our additional network adapter. Currently you can see there is a virtual machine has only one network adapter, the network adapter bound to our port group, our port group,

09:00-09:55
which is called PortGroup-10Net, this is the interface that the management of the device will be used and also for our log forwarding. For Network Traffic another adapter is required. We're going to go back in to our **Add network adapter**. Select the port group previously created that has a SPAN in promiscuous mode configured from the drop-down section, and then choose **Save**. Now we have two network interfaces for this Modular Sensor. It is now ready to be powered on.

The lesson VMware ESXi configuration is complete. You should now be familiar with how to create a new ESXi port group. Thank you.

L03B-DEMO: ESXi Configuration (11m:50s)

00:00-01:00
The next lesson in the *Stellar Cyber Modular Sensor Installation and Configuration* module is *VMware ESXi Configuration. VMware ESXi Configuration.* This demonstration includes the ESXi configuration and networking configuration, including virtual switches, virtual port groups, and the additional network interface required for the network traffic monitoring for the Stellar Cyber Modular Sensor. We'll also demonstrate creation of a Modular Sensor virtual machine and how to edit a virtual machine to add an additional network interface.

Let's start with the networking configuration in our ESXi box. Okay. Click on **Networking**. We'll start here with the virtual switches. Here's what we can see.

01:00-02:00
The ones we're going to use today are–we have vSwitch0, which is the default that comes with ESXi. But we've already created a virtual switch-10Net. That virtual switch is what we'll use for our ten network. We'll see we're using the 10.20.1.0 network. That's our virtual switch. If we look at the interfaces that are connected, currently we have two port groups. We have a PortGroup-10Net or PortGroup-10Net. That's what we're going to be using for our management interface and our log forwarder interface on the Modular Sensor. As we can see here, connected to the virtual switch-10Net that we just talked about. Okay. For the port

groups, we need an additional port group. The first thing we have to do is go up and add a port group. Let's call this port group, "PortGroup-NDR-SPAN-10Net."

02:00-03:00
This will be the port group that will collect the SPAN traffic. Now for a VLAN ID, we need a VLAN ID of 4095. What that ID is for it's a particular VLAN ID for VMware ESXi. What it's saying is that the devices that are connected to this port group are able to accept tagged and untagged ethernet frames. If we're talking about VLANs and 802.1Q frame tagging or VLAN tagging, we're able to have an interface and the Stellar Cyber Modular Sensors interface is able to accept tagged and untagged frames by calling this VLAN ID on this port group 4095.

03:00-04:00
That's how we identify to VMware that the devices connected to this port group are able to accept tagged and untagged frames. In this example, we won't have any tag frames. An example of a tag frame would be if you had a physical port connected to a trunk interface on a switch that was carrying multiple VLANs, and those multiple ethernet frames with multiple VLANs would be identified by their VLAN ID or that 802.1Q tagging information. That's how this is to be understood in VMware. Now it's not connected to vSwitch0. We want to connect it to vSwitch 10.net. Then there's one more thing we need to do. We want this interface to act more like a hub interface. We want to see all of the traffic in this virtual switch.

04:00-05:00
Rather than a normal switch where it builds a MAC table based on ports that it builds from learning source MAC addresses, we want this port to accept all frames in the switch regardless of what that MAC table looks like. The way we do that is we click on **Security** and we turn on **Promiscuous mode**. In this example we would click **Accept**. There is a different way you could do it. You could actually turn on promiscuous mode at the virtual switch level. If we had turned it on–move this out of the way–we could actually turn it on at the vSwitch-10Net. You could turn on promiscuous mode, in which case we wouldn't have to do it for this, but it's nearly like saying we're going to turn it on for everyone and we can choose who to turn it off for. But for our configuration, it is not the virtual switch that is behaving like a switch. For this interface, we need to turn on promiscuous mode, accept, and then click **Add**. Now we'll see we have our two network ten interfaces.

05:00-06:00
We've got our PortGroup-10Net and we've got our PortGroup-NDR-SPAN-10Net. We can see here that it is VLAN 4095. There's nowhere in this table where it shows that promiscuous mode, but we know that we've just configured that. The next thing to do is create the

Modular Sensor virtual machine. To do that we click on **Virtual Machines**. At the top, **Create / Register VM**. Select that. We're going to create this virtual machine from an OVA file that has been previously downloaded from Stellar Cyber. Deploy a virtual machine. We're going to use an OVA file. Select **Next**. We give the virtual machine a name: "SC Modular Sensor." We can either drag the OVA file into this blue space or we can click to browse.

06:00-07:00

I will click to browse in this example. We have put it in the documents directory. There is our file: our aella-modular-ds-4.3.7.ova. Note this is a 1.2-gig file, and part of this process will have to copy that file from this local machine to ESXi. That takes a little bit of time depending on your network bandwidth. Right? Now we've given it a name. We've got the file the OVA file selected. We select **Next**. This is where we will store this virtual machine. We're going to store this virtual machine on our Dell RAID 2 Data Store. The only one available to select and there is plenty of free space. Select **Next**. Okay. There's a couple of things to check here. Number one, the VM network, this PortGroup-10Network, is actually correct.

07:00-08:00

It's selected the right one. We can see they're the ones we had to choose from the 192 network. We're certainly not putting this virtual machine on the 192 network. The SPAN port group we configured before is for the network monitoring, the network traffic. This is actually the correct one: PortGroup-10Net. We want **Disk provisioning** to be **Thick**, which just means that the disk is allocated ahead of time rather than expanded as needed. We're going to uncheck the **Power on automatically** because we have some configuration changes we'll make to this virtual machine before we power it on. Next one is select **Next**. Here's our summary of our virtual machine that's ready to deploy. As soon as we click **Finish**, it will start copying that OVA file and building our virtual machine.

08:00-09:00

Remember we unchecked **Automatically power on**, so we'll have the opportunity to edit it before we power on. Let's select **Finish**. We look down at the bottom in these tasks. We'll see a couple of tasks get created, which is copying that OVA file from our local host. There you go. We can just see that Stellar Cyber Modular Sensor has been created as a virtual machine. The one we're looking for here is this upload disk Stellar Cyber that's our OVA file. We want to watch this through to completion. We can see it's running at up to four percent. We'll wait till it's 100 percent complete and continue with editing our virtual machine before powering it on.

09:00-10:00

We're just about finished at 98 percent. That's uploading that OVA file to ESXi, storing it in our Data Store. We've got a couple more configuration items to do. We'll just wait for this to

finish the final piece. Here we go. Okay. Completed successfully–that's what we want to see, if you want to know why there's two line items here, you can see we've got an upload disk. That's the one we just did that was completed successfully. This one which has a result of failed upload disk canceled. I actually forgot to press **Record** halfway through configuring our Modular Sensor, so I have to stop and start again. Okay. Select our Stellar Cyber SC Modular Sensor, and we're going to make some edits to this virtual machine.

10:00-11:00
At the moment we can see over here we've got four virtual CPUs, eight gig of RAM, 64 gig of SSD hard disk, one–only one–network interface connected to PortGroup-10Network. Our 10.20.1 network is where it's going to be connected to. The first thing we have to do is edit that and add our additional network card. We choose **Edit**. We can see here we only have one networking card. At the top to say **Add network adapter**, we've now got two network cards. Both of them are connected to the same port group. We need to select our SPAN port. That's that one there, and that is that configuration done. Select **Save**, and we are ready to power on, so **Power on**. I will open up the console so you can see it booting.

11:00-11:50
There's our login prompt. We are ready to configure the Modular Sensor.

The lesson *VMware ESXi Configuration* is complete. You should now be familiar with how to create a new ESXi port group. Thank you.

L04 - MODULAR SENSOR CLI CONFIGURATION

L04A-PRES: Modular Sensor CLI Configuration (05m:20s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Modular Sensor Installation and Configuration* is *Modular Sensor CLI Configuration. Modular Sensor CLI Configuration.* In this lesson, we take a look at how to configure the Modular Sensor from the command-line interface.

We've configured our Modular Sensor virtual machine in ESXi with two network interfaces and now we are powering on that device. After you've powered it on, open the browser console, and select that little thumbnail for the virtual machine console, and you can watch the Modular Sensor virtual machine boot. The boot sequence will culminate when you see the login prompt. That is good news for us to see the login prompt.

01:00-02:00

From the login prompt, we're going to sign in for the first time. The username is aella, a-e-l-l-a, and the default password is changeme, c-h-a-n-g-e-m-e. Enter **aella** and **changeme** you'll immediately be asked to change the password. The current password is **changeme**. Enter **changeme** and then enter your new password and confirm your new password. With the new password created. It's time to add some IP networking to our Modular Sensor. The first thing to do is to set the interface management IP address. We do that with a `set interface management IP`, and then the IP address followed by a forward slash and the number of bits of the Netmask. In this example, it is `set interface management IP 10.20.1.15/24`.

02:00-03:00

A bit of a trick with the CLI: If we type `set int` and press tab, it will finish the interface command. It'll say `set i-n-t` tab It'll say `set interface`. If we do `set interface m-a-n` tab, it'll fill out "management," so a bit of a quicker way to expand out all of those commands. Next is to set the gateway. The gateway command is `set interface management gateway` and the IP address of the gateway for the Modular Sensor. The final IP address task is to give it a DNS server. In this example, set interface management DNS to 8.8.8.8. To confirm that if we do a `show interface`, we can see in the output from the `show interface` command we can see our IP address there of 10.20.1.15 with a mask of 2552552550, which is the same as a slash 24-bit mask. A `show gateway` command will confirm what we entered before with our 10.20.1.1 in our example.

03:00-04:00

A `show DNS` of `8.8.8.8`. We can see that we have configured that correctly. The very first time we put the `set interface` command in, there was a response that said please restart system for changes to take effect. With all of the configuration done, we're going to restart the system. The command there is `restart system`, and we'll see that virtual machine reboot again. Continuing our configuration of the Modular Sensor, we're going to add a host name and bind that Modular Sensor to the Data Processor using the DP's management interface IP address or fully qualified domain name. These commands are `set host name`. That string that you enter here, that is the name of this Modular Sensor.

04:00-05:00

It will show up in our user interface in the table of Modular Sensors. Choose a name that's meaningful to you. If you have many sensors configured in your network, whether they're Modular Sensors or server sensors, this will be a table with a list of sensors. This name is how you will identify this Modular Sensor. To bind this Modular Sensor to our DP, we do a `set cm`, a `set cm, an IP` address or a `set cm` and a fully qualified domain name. This

is the IP address of the management interface of our data processor. Confirm the host name and connection of the Modular Sensor to the Data Processor with the following commands: A `show version` will show us our host name in that output. A `show cm` will show that we have an established connection to our configuration manager or which is our IP address of the Data Lake Management interface.

05:00-05:20
The lesson, Modular Sensor CLI configuration, is complete. You should now be familiar with configuring the Modular Sensor from the command-line interface. Thank you.

L04B-DEMO: Modular Sensor CLI Configuration (08m:20s)

00:00-01:00
The next lesson in the *Stellar Cyber Modular Sensor Installation and Configuration* module is *Modular Sensor CLI Configuration. Modular Sensor CLI Configuration*. In this demo, you will see how to configure the CLI configuration, including the initial login with a password change, setting the Modular Sensor network configuration, verifying the network configuration, and restarting the system. We'll also configure the host's name and configuration manager settings.

Alright. First thing, let's find our virtual machine. We select from the left-hand panel **Virtual Machines**. Down at the bottom is SC Modular Sensor, the one we configured in the last lesson. We open that up. Look at the console.

01:00-02:00
We can see we have a login prompt ready. Our username is **aella** and the password is **changeme**, c-h-a-n-g-e-m-e. It's asking for the current password again because we need to set a new password. It's asking again for changeme and now it's asking for the new password. Alright. I've changed the password and I have logged into our Modular Sensor. The next thing we need to do is set up the networking. First things first, `set` in this UI in this console, we type in enough of the command to make it unique. I can press tab to complete it. The command we're going to look for to give it an IP address is `set` it said interface, `i-n-t`. I press tab and it will spell it out for me.

02:00-03:00
It's `set interface management IP`, `set interface m-a-n` press tab, spells out management for me, IP, and the address we're going to give it is 10.20.1.15 with a mask of /24. **Enter**. It says here, "Restart system for changes to take effect." We're going to complete

all of the IP networking configuration, and then we will restart the system. We've given an IP address. That's the IP address that we'll use for our log forwarding address and also an address that we can use to manage the device. We could SSH to this device using that 10.20.1.15 address. The next thing we need is to give it a gateway, `set int`. I press tab, `man` tab, `gateway` tab, space and our gateway is 10.20.1.1. **Enter**.

03:00-04:00

We have to give it a DNS address. We `set interface`. Tab `management`, tab, `dns`, `8.8.8.8`. That is a Google DNS server. **Enter**. Alright. Now that we've done all of those things, let's restart the system. The first thing we'll check when the system restarts is that we have got our configuration correct. There's a couple of show commands we can use to confirm our configuration. We're just about finished booting. There's a login prompt, aella. This is not changeme anymore. This is our new password.

04:00-05:00

Okay. There I am. I'm in. The first thing we can do is `show interface`. We can see our IP address: 10.20.1.15. That's correct with the mask of 255.255.255.0. Or we could say that is a /24-bit mask. Our interface IP addressing is correct. The next one to check is `show gateway` to see if we've got that configured correctly. That is configured correctly. The last one is a `show DNS` to make sure that it's configured. How could we test this to make sure it is actually all working? The first thing we could do is ping the gateway, so `!ping 10.20.1.1`. Great. That is working. A Control C to stop. Then the next one we might do is `ping 8.8.8.8`.

05:00-06:00

That will prove that our networking is working, that our default gateways are going. Right? We're out to the internet that's working. Control c to stop that. The last one, let's make sure that our DNS is working. A couple of different ones here we could do: a ping to Google.com. You can see here our ping response is working. Control c to stop. Another one we could do a `shell dig google.com`. We can see there we've got the right information coming back. All of our IP addressing have been configured correctly. The next thing I'm going to do is give it a host name. This host name will show up in the DP UI to identify this Modular Sensor. We're going to do a `set host name`. Sorry. I pressed tab there to fill out that command: `set host name` and we're going to do `sc-modular-sensor`.

06:00-07:00

Enter. The most important piece now is to connect or bind this Modular Sensor to our DP, to our Data Processor. The way we do that is with the `set cm` command, `set cm`. The address we're going to use is 192.168.1.175. That is the management interface of our Data Processor. Whatever that address is, and of course it can be a domain name, whatever it is that we use

to connect to our Data Processor, that's the address that we use here. **Enter** to do that. Now we can do a couple of `show` commands to confirm what we've just put in. The first one we can do is a `show version`. We can see in a `show version` a couple of things. The first one I want you to look at is the hostname.

07:00-08:00
We can see that about seven or eight fields down. We see hostname sc-Modular Sensor. We know we've configured that right. We can see that the management IP is correct with a 10.20.1.15/24. We can see our CM is correct: 192.168.1.175. That's what we're looking for there to make sure that we've configured those correctly. There is another command that is more important to show the connection to that CM. That configuration management or our DP management interface is actually working is a `show cm`. The good news there is that what we want to see is that status of ESTABLISHED. We now know that our Modular Sensor has been configured. It is talking to our Data Processor. But there's some configuration we must do on the Data Processor side to accept that connection from this Modular Sensor.

08:00-08:20
That's all we've got for this piece.

The lesson, *Modular Sensor CLI Configuration,* is complete. You should now be familiar with configuring a CLI configuration. Thank you.

L05 -DATA PROCESSOR CONFIGURATION FOR MODULAR SENSOR

L05A-PRES: DP Modular Sensor Configuration (03m:25s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Modular Sensor Installation and Configuration* module is *Modular Sensor DP Configuration. Modular Sensor DP Configuration*. In this lesson, we authorize the Modular Sensor in the Data Processor's user interface.

After configuration of the Modular Sensor virtual machine is complete, the remainder of the configuration is completed into the Data Processor's user interface. The first thing to do is to go to our **System | Sensors** to see our Modular Sensor in the table. We can see with the configuration of the Modular Sensor in the virtual machine complete with the confirmed

"ESTABLISHED" from that `show cm` command, the Modular Sensor is added to the sensor table.

01:00-02:00
Note the red dot under the **Sensor Status** column so that we need to become a green dot and the red lock and "Not Authorized" under the **Authorization** column. One thing we need to do is to make sure that that red dot under the **Sensor Status** comes green and the second thing we need to do is authorize this Modular Sensor to be bound with our Data Processor. The first thing to do to authorize our Modular Sensor, we can if you had multiple Modular Sensors, click the check mark next to them or simply select the **Manage** button from the right-hand side. After selecting the **Manage** button, we get a drop down and we want to do authorization. That's what we're doing. We're binding our Modular Sensor that was created in ESXi, and we're binding that to add this Data Processor. We do that via authorization.

02:00-03:00
When the **Authorization** window pops up we have the choice at the top. You can see there are some radio buttons to either **Authorize** or **Unauthorize** wherein we are creating this so we're going to authorize from the left-hand side. That window shows us the sensors available to authorize. We can see there we've just got that one sensor to authorize so select that from the box on the on the left-hand side of the name and click the **Add** arrow button to add our Modular Sensor to the **Authorized** section. Then the final thing to do is select the **Submit** button below to complete that authorization. Now the Modular Sensor is authorized. We can see our status is now green and our **Authorized** column has now a green shield, and the label, "Authorized." You may have to wait a moment or two before that happens. That green dot doesn't come there straight away; you may have to refresh this table while that authorization happens to get our green status dot.

03:00-03:25
Now as we said before, we're looking for our green shield and "Authorized" under that **Authorization** column.

The lesson *Modular Sensor DP Configure* is complete. You should now be familiar with how to authorize the Modular Sensor in the Data Processor's user interface. Thank you.

L05B-DEMO: DP Modular Sensor Configuration (05m:00s)

00:00-01:00

The next lesson in the *Stellar Cyber Open XDR Modular Sensor Installation and Configuration* module is *Modular Sensor DP Configuration. Modular Sensor DP Configuration*. In this lesson, we authorize the Modular Sensor in the Data Processor's user interface.

In this demonstration, I will show you the process of how to authorize our Modular Sensor through the UI of the Data Processor. To do that, we select from the menu **System | Sensors** under the **Collection** column. We can see here we have our Modular Sensor, SC-modular-sensor, which is the name we gave it when we configured it, configured the host name in the CLI of the Modular Sensor. I can just expand that column so we can see the entire name. There we are. Alright.

01:00-02:00
No messages. The great news is we have a big green light under the **Sensor Status** column. I'll also expand that column so you can see everything. What the green light is showing us, if I go back to our console of the sensor. Log in here. That green light in our **Sensor Status** profile is coming from us doing a `show cm`. That "ESTABLISHED" there is what's showing us the green light under **Sensor Status**. Back to our sensor. What we're seeing here is "Not Authorized" under the **Authorized** column. Again, I'll expand that out. Authorization not authorized. We've got a red lock meaning that we've connected our Modular Sensor to the Data Processor, but it's not yet authorized.

02:00-03:00
Here's how we go about that. If I select the **Manage** button up here, choose **Authorization**, we get a pop-up window. In the left-hand side panel, we can see our sensor, sc-modular-sensor, and there's a checkbox next to that. We can either choose just to select that one or if we had many that we were doing at the same time, we could choose **Select All**. Once that's selected, we select the **Add** button to move our SC Modular Sensor over to the **Authorized** section and all that's left to do now is to select the **Submit** button. What will happen now you can see that the **Sensor Status** button is now red. What is happening is a configuration is being pushed down to our Modular Sensor based on the profile that it's been associated to.

03:00-04:00
By default, our Modular Sensor will be associated and we can see over here if we scroll over, to the **Sensor Profile**, we can see it's been associated to the Default_modular profile. If we were to look at that you can select it right from here. Let's do this while we're waiting for our **Sensor Status** to complete and our Modular Sensor to reboot with that new configuration that's been pushed to it. I select **Default_modular** and we're going to have a look in here. As we said previously, Log Forwarder is on by default. We can't turn that off.

Here are our other available features to turn on for this profile. We would turn those on and select **Submit**. We will do those in a later lesson. This is the profile that this sensor has been bound to and the configuration in that profile is what's been pushed down to our sensor. We can see our sensor has come back online. It is green. We can see success. It is now authorized.

04:00-05:00
We can see that with our green shield. If I now go look at the details of our sensor by clicking on the name, we can now see that it is authorized down here. The status is green. Over here it's healthy. It's got its configuration, the control plane is working. The one we're most interested in here is the Log Forwarder service is working. Our **Sensor Status** is green and has been authorized. If we go back to our list we'll see that is all the case. So that's how you authorize a sensor in the UI of the Data Processor.

The lesson *Modular Sensor DP Configuration* is complete. You should now be familiar with authorizing a Modular Sensor through the user interface of a Data Processor. Thank you.

L06 -MODULAR SENSOR CONFIGURATION VALIDATION

L06A-PRES: Modular Sensor Configuration Validation (05m:35s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Modular Sensor Installation and Configuration* module is *Modular Sensor Configuration Validation. Modular Sensor Configuration Validation*. In this lesson, we validate the Modular Sensor and Data Processor configuration by looking at the syslog traffic arriving to both the Modular Sensor and the Data Processor.

So far we've created our virtual machine, our Modular Sensor in ESXi. We've given it two networking interfaces: we've given it a LAN interface or a management interface, and we've given it the SPAN interface in that configuration. In the Data Processor user interface, we have authorized that Modular Sensor to communicate. It's been bound to our Data Processor or our Stellar Cyber platform.

01:00-02:00
What I want you to pay attention to here is there's a firewall located between the virtual switch in ESXi. It's a pfSense. We can see we've got a firewall that sits between our vSwitch LAN and our vSwitch WAN. The firewall was previously configured to send syslogs to

10.20.1.15. That's our Modular Sensor. It was configured to send on port 5543. We're going to validate on both the Modular Sensor and the Data Processor that the logs from that pfSense firewall are arriving to our Modular Sensor at 10.20.1.15 on port 5543. If you go into our documentation, on the Knowledge Base you can see that port 5543 is the parsing port for a pfSense firewall.

02:00-03:00

That Modular Sensor will be able to parse the logs that are arriving and then normalize and forward to our Stellar Cyber Data Processor. We're then going to validate on the Data Processor that those logs have arrived. Let's confirm on the Modular Sensor console or in the CLI of our Modular Sensor. If we do a `show cm` or `show configuration manager`, what we're looking for is that the CM controller is 192.168.1.175. That is the IP address of our Data Processor, the Data Lakes Management IP address. We can see there that it is established. The keyword is the status is ESTABLISHED. At the other end, on the Data Processor, we went through the process of authorizing this Modular Sensor to send traffic and communicate. The next thing we can do is a `show logforwarder`.

03:00-04:00

The data under the input shows the logs received and we can see under **Input** we have pfSense_fw, and the number is the number of logs that have been received. The data under output shows what was forwarded to the Data Processor. We can see here that on the Modular Sensor, logs are being received from the pfSense firewall and they are being forwarded to our Data Processor. If we go into the Data Processor for validation, so the first thing to do here we can see we are in **System | Sensors** looking at the **Sensor** table. We've got a green **Sensor Status**. It's been authorized. We select the Modular Sensor by clicking on the name to open up that pop-up window for this Modular Sensor. The first thing we can see there in the Modular Sensor we have two counters going on. We're going to look for the **Log Forwarder Input** and the **Log Forwarder Output**, and to see those counters increasing.

04:00-05:00

Now this is a different screenshot from a different time. We've got more than those couple of records that have been coming in. There are many, many different log messages coming in in this graphic. Opening up **System | Sensors**, selecting your Modular Sensor, will pop up this window where we can see those numbers increasing. This is another way to validate that our configuration is correct and that pfSense firewall that is sending to our Modular Sensor that those logs are arriving, they are being parsed. Then they'll be normalized and forwarded to our Data Processor. Another thing we can look at in the Data Processor UI. If we go to the **Investigate | Threat Hunting** page and choose the **Syslog** index, so by default

when we go to I**nvestigate | Threat Hunting** it will be the **Alerts** index. We're going to change that index from **Alerts** to **Syslog**.

05:00-05:35
We can start to see that we are seeing data in this page from the pfSense firewall. We'll see that they're under one of the log classes that we see pfSense_firewall. This is yet another way to validate the configuration of our Modular Sensor, both in ESXi and in the Data Processor have been done correctly.

The lesson, *Modular Sensor Configuration Validation,* is complete. Thank you.

L06B-DEMO: Modular Sensor Configuration Validation (09m:40s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Modular Sensor Installation and Configuration* module is *Modular Sensor Configuration Validation. Modular Sensor Configuration Validation*. In this demonstration, we'll validate the authorization of the Modular Sensor in the user interface of the Data Processor.

In this demonstration, we will validate the authorization of the Modular Sensor in the UI of the Data Processor. We'll validate that by looking at log forwarding from the firewall. We'll look at that on the Modular Sensor. We'll validate the logs have been received by the Data Processor, looking at Modular Sensor counters inside the UI, and we'll also validate those logs in the Data Lake by looking at **Investigate | Threat Hunting**.

01:00-02:00
The first thing we'll do is jump into the console of the Modular Sensor. (types) Aella. Here we are. Okay. The first thing to check, `show cm`, that ESTABLISHED that we see there, that is what gives us our green status symbol for our Modular Sensor online. That's great news. We expected to see that–to see logs coming into our Modular Sensor. We will do a `show logforwarder`. That's all we've got to type. Press tab to complete the command. Enter. What we can see here is as I said before, we have two preconfigured firewalls on this network. When I say preconfigured, they have already been configured to send their log traffic to our Modular Sensor on 10.20.1.15.

02:00-03:00
Just to show that, we do a `show interface`. There it is: 10.20.1.15. Those two firewalls, a pfSense firewall and a Sophos firewall, have previously been configured to send their log traffic to this Modular Sensor. That's why we can see under input pfsense_fw, at 429, 429 logs have been received by the Modular Sensor so far. We can see the Sophos firewall has

sent 180 logs so far. That's one way we can confirm that log traffic is being received by the log forwarding engine on the Modular Sensor. What we're going to do now is go into the UI of the Data Processor to confirm what we're going to see in there of that log traffic arriving. To confirm log traffic arriving in our Data Processor.

03:00-04:00

Here we are in the UI. The first place we can go and look is the Modular Sensor. From the menu **System | Sensors**. Here's a table of our sensors, all of them, all of our one of them sc-modular-sensor. If I select that Modular Sensor, we've seen this page before with our green authorization, green status, and green log forwarding service. That's all fantastic. What I want you to pay attention to is the **Log Forwarder Output**. Although the title is output, there's two numbers in there, an input and an output, so those numbers are confirming that log traffic is being received by the Modular Sensor and has been sent by the Modular Sensor to our Data Lake. You can see if we continue to look at those numbers we'll see those increasing. If we go further down, and click on our **Log Forwarder stats**. What we can see here is our **Log Forwarder Input** rate over time.

04:00-05:00

What we're seeing is the traffic, the log traffic arriving over time. If I look at the **Log Forwarder Summary**, what this is showing us is we have two of the top 20. We've got two IP addresses that are sending log traffic to this Data Processor. Source IP address of 10.20.1.1. You can see that sent 60,000 or responsible for 60,000. That's a bit of a split between the blue and the green of the 60 versus the 106. 10.20.1.1 is also the default gateway of the Modular Sensor, which happens to be our pfSense firewall. We can see on the right-hand side of that table, that port 5543 is what's selected there. Port 5543, if you look in the documentation for parser ports, you'll see that that is the port to forward traffic from a pfSense firewall

05:00-06:00

to a Modular Sensor so that when the traffic arrives, the Modular Sensor knows how to parse that. We know what the formatting of that log traffic looks like, we understand the field names, we understand the value formatting so that we can parse that traffic and normalize it into an Interflow record. The second one, which is green on the left, is 10.20.1.99, I know I can tell you that is the Sophos firewall that is also sending traffic, and it's sending traffic to 5520. That's the port that you would find again in the documentation for parser ports. That's the port number that will parse log traffic from a Sophos firewall. Now that's one place we can look at this log traffic and confirm that it's coming in. The second place we can at that is in **Investigate | Threat Hunting**. Let's do that now. We select from the menu **Investigate | Threat Hunting**.

06:00-07:00
By default, we can see that the **Alerts** index is chosen. The **Alerts** index is always the default index that's chosen whenever we go in to **Investigate | Threat Hunting**. So we need to change that index from **Alerts** to **Syslog**. The way we do that is select **Alerts**, or the button that happens to have **Alerts**, uncheck **Alerts**, and check **Syslog**. Now all we're looking at are log entries in the Syslog index. We can see if we look at the **Top Log Sources** that we can now see the purple is fw_sophos, and the gray is pfsense_fw. There are two firewalls that are sending traffic to our Modular Sensor. Let's filter out this Stellar Cyber one and just remove that so we're looking at our two firewalls. We're looking at the two firewalls: pfsense_fw and fw_sophos,

07:00-08:00
looking at the **Top Log Source** classes, you can see there's only one. The only class for this DP of logs that have been received is the firewall one. If we go scroll further down into the table, at the moment that table just seems to be full of pfSense firewall, but right there at the bottom we can see some, a couple of entries for the Sophos firewall is also sending log entries. That is the way we can validate that log traffic is being sent to the Modular Sensor on particular port numbers so that they can be parsed correctly. Those logs are then normalized and enriched. We'll talk about that in another lesson. Forwarded to our Data Lake, and we can go and see those counters under the Modular Sensor and we can see the actual logs arriving. We can look at one if you'd like. Let's open one and have a look. We'll go to the **Details**.

08:00-09:00
Here is the log coming in. We can see that the engine ID is modular. That's how the traffic that's how the log has been received. The device is a pfSense firewall. What else can we see? The message and origin category is firewall, the class is firewall, the message origin source. This is a field name that is used often when trying to find out what is the origin of this entry. You can see there it's a pfsense_firewall. The address in which it arrived at: 10.20.1.1. That's where it's coming from. Alright. The only other one there I'd like to point out is the stellar_ingestion_port, that 5543. That is the port number we expect to see pfSense firewall, log traffic on so it can be parsed correctly.

09:00-09:40
There you go. That's how we can confirm that our configuration of our Modular Sensor is actually working for log forwarding traffic. The next piece we need to do is talk about how do we turn on the Network Traffic monitoring, because at this stage all we're seeing is log forwarding traffic. That is the end of our demonstration.

The lesson, *Modular Sensor Configuration Validation,* is complete. You should now be familiar with validating a Modular Sensor through the user interface of a Data Processor. Thank you.

L07 -MODULAR SENSOR TRAFFIC MONITOR CONFIGURATION

L07A-PRES: Modular Sensor Traffic Monitoring Configuration (04m:05s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Modular Sensor Installation and Configuration* module is *Modular Sensor Traffic Monitoring Configuration. Modular Sensor Traffic Monitoring Configuration*. In this lesson we add the Network Traffic feature to ingest network SPAN traffic.

In this network diagram, our Modular Sensor has two interfaces. It has a management interface that is also used for log forwarding, and it has a Network Traffic interface that is used for our Network Traffic analysis. We can see there inbound to that virtual switch LAN is port group SPAN and port group LAN. They are the two port groups that are bound to our Modular Sensor. This lesson will configure the Modular Sensor via the sensor profile.

01:00-02:00
We don't edit the Modular Sensor itself. We edit the profile that the sensor is bound to to turn on ingestion of that Network Traffic so to enable the SPAN traffic in our orange interface there to bring in traffic into our Modular Sensor. Number one, let's just review our Modular Sensor virtual machine. We can see here we have two port groups attached to our two network interfaces. We have PortGroup-10Net and we have PortGroup-NDR-SPAN-10Net. The PortGroup-10Net is bound to our network adapter number one. Our PortGroup-NDR-SPAN-10 is bound to network adapter number two. The first one is what's used for our management and the second one is what's used for our SPAN traffic. That's confirming what we configured earlier.

02:00-03:00
If we go into the console, we'll see that a `show interface` command only shows one interface we're seeing the management interface here. At this stage we don't have our second interface or an additional interface configured in our profile. Let's go see how to configure that. Within the UI, it's not the sensor that we configure. It's the sensor profile. If we navigate to **System** and **Sensors** we see a table of our sensors available. We're looking at

the sensor named sc-modular-sensor. Along that line we can see that it has a sensor profile called Default_modular. Now we recommend that you actually use a different profile depending on your requirements. But in this example, our Modular Sensor here is using Default_modular for its sensor profile.

03:00-04:05
It's the sensor profile that requires the configuration change to add the Network Traffic, so select Default Modular. There's a button that we can select to edit that sensor profile. Our **Sensor Profile** window will open for editing. We can see here by default, the log forwarder is the only feature enabled. To enable the Network Traffic feature in this sensor profile, select the slider to the left of Network Traffic, and turn it on. You can see there that it turns on and a bunch of sub features are also turned on for Network Traffic. Once you have done that, scroll to the bottom and select **Submit** to turn on that Network Traffic feature for this sensor profile that our Modular Sensor is bound to. That's that connection piece that we're trying to get there.

The lesson, *Modular Sensor Traffic Monitoring Configuration* is complete. Thank you.

L07B-DEMO: Modular Sensor Traffic Monitoring Configuration (05m:35s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Modular Sensor Installation and Configuration* module is *Modular Sensor Traffic Monitoring Configuration*. *Modular Sensor Traffic Monitoring Configuration*. In this demonstration, we'll show you the Network Traffic configuration including confirmation of the ESXi configuration of the Modular Sensor virtual machine.

In this demonstration, I will show you the Network Traffic configuration, including confirmation of the ESXi configuration of the Modular Sensor virtual machine, particularly looking at the interfaces configured. We will look at Modular Sensor to show the interface output, so look at the console to see what that looks like. Then we will enable network traffic on the sensor profile that the Modular Sensor is bound to.

01:00-02:00
Let's jump in and have a look. First thing we will look at is ESXi. Here is our Modular Sensor, SC Modular Sensor. We can see over in the **Hardware Configuration** we have two network interfaces: Network adapter 1, which is attached to PortGroup-10Net, and we have Network adapter number 2, which is attached to PortGroup-NDR-SPAN-10Net. They're both

connected to the same network. What we mean by that, If we actually look at one of these, let's select the first one This interface, PortGroup-10Net, is attached to the vSwitch-10Net. You can see there are a bunch of virtual machines whether Active Directory servers or Linux servers or CentOS servers all connected including our Modular Sensor connected to this virtual switch.

02:00-03:00

All of that traffic is just regular network traffic attached to that switch. If we go back to our Modular Sensor and look at the other port group, which is attached to network interface number two, we'll see it's also connected to vSwitch-10Net. The difference between this interface, if you remember, if we go and edit that interface we'll see that under the **Security** settings we have **Promiscuous mode** set to **Accept** meaning it will receive all ethernet frames inside of that virtual switch. Any traffic in vSwitch ten going from host to host, including those Active Directory servers, those Linux servers to and from the firewall will all be seen by this interface: PortGroup-NDR-SPAN-10Net.

03:00-04:00

That's what we see. Now we want to see what that looks like from a configuration perspective on our Modular Sensor. Alright. Let's log in to the console. (Enters) aella, and our password. Now if we do a `show interface` to look at what interfaces we've got, we can see there we've only got the management interface and we said in the previous lessons. This interface is the way we put SSH to this interface. It's a map from a management perspective. It is also the interface that our log forwarding traffic arrives on so that 10.20.1.15. We can see in the background of this screen, the configuration of the actual virtual machine has two network interfaces–Network adapter 1 and Network adapter 2 both connected–but here we're only seeing one.

04:00-05:00

The reason for that is at this stage we haven't got **Network Traffic** turned on in the profile that this Modular Sensor is connected to. Let's turn that on. Back to the UI. We're looking at our interface here. We've got our Modular Sensor. It is connected to sensor profile default modular. If we look over here under our services, we can see there's no Network Traffic. Everything else is green, but no network traffic. To configure that let's go edit the sensor profile, and it's simply a matter of turning on the switch for **Network Traffic**. All of these are on by default. We scroll down to submit. We've now edited that sensor profile. What will now happen is the new configuration for any sensors that are bound to that profile will have that new configuration push down.

05:00-05:35

If we go ahead and look at **Sensors**, we'll probably see a big red dot as we do, because that sensor is in the middle of updating its configuration because we've changed the configuration in that sensor profile. We'll wait for that to happen and come back and do some validation.

The lesson *Modular Sensor Traffic Monitoring Configuration* is complete. You should now be familiar with network traffic configuration. Thank you.

L08 -MODULAR SENSOR TRAFFIC MONITOR VALIDATION

L08A-PRES: Modular Sensor Traffic Monitoring Validation (03m:45s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Modular Sensor Installation and Configuration* module is *Modular Sensor Traffic Monitoring Validation*. *Modular Sensor Traffic Monitoring Validation*. This lesson focuses on validating the configuration of the **Network Traffic** feature by looking at the Modular Sensor CLI and the Data Processor user interface for increases in counters and seeing the records in the **Investigate | Threat Hunting** page.

Now that we have made our change to the sensor profile that is bound to our Modular Sensor, and we've included network traffic in that sensor profile configuration. That configuration has then been pushed down to our Modular Sensor and we can now see here with a `show interface` command that we have more interfaces, more than just the management interface that we had before.

01:00-02:00
We had a management interface that is for managing the device and for our log forwarding. We can now see we have additional interfaces. I want you to pay attention to our ethernet1 interface. This is the interface for ingesting network traffic in our Modular Sensor. Run the `show interface` command a number of times. What I want you to pay attention to is look at the RX bytes for the interface ethernet1. The volume should increase in response to the network traffic. If there's traffic that is occurring on that network segment, because we have turned on promiscuous mode in the virtual machine from ESXi, that's saying I want to receive all frames that the interface is connected to in that virtual switch.

02:00-03:00

If there are other devices communicating, then they will also be forwarded to this interface and we should see the RX bytes increase. This is one way to validate that our traffic monitoring is working at the sensor level. In the Data Processor's user interface, navigate to **System** and **Sensors** to show the sensor table. We can see here our Modular Sensor, sc-modular-sensor, is listed. What I want you to pay particular attention to is the **Input** and **Output** column values. This is another place where we can validate the ingestion of that network traffic. We could first see it in our Modular Sensor console. We're now seeing that reported in the UI of our DP. Another place to validate it is in the **Investigate | Threat Hunting** page. We did the same validation with our log traffic. We're now doing this in the UI for our network traffic.

03:00-03:45
Again, the default for **Investigate | Threat Hunting** is the **Alerts** index. We need to change that **Alerts** index to the **Traffic** index and we can now see that we have traffic being ingested from our modular_sensor. That is the type of sensor that traffic's coming in from. We validated it coming into our sensor. We validated it in our sensor **Details** page from our **System Sensor** table, and now we're validating that traffic coming into our Data Lake as records, Interflow records, for network traffic.

The lesson *Modular Sensor Traffic Monitoring Validation* is complete. Thank you.

L08B-DEMO: Modular Sensor Traffic Monitoring Validation (06m:25s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Modular Sensor Installation and Configuration* module is *Modular Sensor Traffic Monitoring Validation. Modular Sensor Traffic Monitoring Validation.* This lesson demonstrates validating the configuration of the Network Traffic feature by looking at the Modular Sensor CLI and the Data Processor user interface for increases in counters and seeing the records in the **Investigate | Threat Hunting** page.

In this demonstration, I will show you the network traffic validation, including checking the Modular Sensor console to look at the interfaces, validating an increase in network traffic statistics in the Modular Sensor console, again using the `show interface` command, validating an increase in the network traffic from the Data Processor side,

01:00-02:00
so looking in the UI at the Sensor table, and validating traffic in **Investigate | Threat Hunting**. Alright. Let's start by checking the Modular Sensor console looking at the interface

settings. Let's just look at our console in ESXi. Previously, we saw we had one interface. Let's have a look now. (enters) aella. Okay, a `show interface`. Now we have four interfaces. The ones we're interested in clearly still in the **management** interface. That's the 10.20.1.15 interface, how we would SSH to it or where our log forwarding traffic is sent to. Any devices, as we saw before, we have our firewalls as sending their log traffic to that interface: 10.20.1.15. The interface we're looking at for network traffic is ethernet1. We'll see a little further on.

02:00-03:00

We're particularly interested in those RX bytes that you can see is currently 25.6 megabytes. I'm going to go to another host that's on this network, one of those active directory hosts, and just run a speed test to generate some traffic. Stand by while I go generate that. Okay. I have just gone on to one of the hosts on this network, this 10.20.1 network, and ran a speed test. It's a tool to generate a bunch of traffic. If I press the up arrow and enter what we're looking for is ethernet1 to see if the counter has gone up for RX. If we see at the top part of the screen, the RX traffic was 25.6 megabytes. We look at the second `show interface` command for ethernet1, we can see the RX bytes is now totaling 1.4 gigabytes. We are seeing network traffic from the Modular Sensor's perspective.

03:00-04:00

Remember, if we look on the right-hand side of the screen, that Network adapter 2, the PortGroup-NDR-SPAN-10Net, is seeing that traffic because of that promiscuous mode. If that wasn't turned on, the switch being a switch, the traffic wouldn't be seen. It would be ignored by that interface, but because it's in promiscuous mode it is seeing the traffic going from the Windows server host to the firewall and out to the internet to do that speed test. It's just a way for us to generate some traffic. Alright. Now let's go into the Data Processor UI and see what that looks like. Okay. Here we are at the home screen. What we're going to look at is first from the sensor perspective, **System | Sensors**. The very first thing we can see here in this table for our Modular Sensor is the **Input** and **Output** columns.

04:00-05:00

We can see that they have increased. Previously they were zeros when we were just looking at log forwarding traffic that was coming in. They were zeros, but now we're looking at Network Traffic we can see the **Input** counter is growing and the **Output** counter is growing. If we go into the Modular Sensor itself and look at the **Details** page, we can see there Interflow stats. Input 1.5 gigabytes and output of 2.5 megabytes of output there. That shows us that network traffic monitoring is enabled. We can also see in this table that network traffic, or sorry excuse me, traffic monitoring under the service has a status of green, so that is monitoring the traffic on that local interface. The next place we can go validate that

network traffic is coming in is under **Investigate | Threat Hunting**. Let's go there. We know that by default, we're looking at the **Alerts** index.

05:00-06:00
Whenever we go to **Investigate | Threat Hunting**, by default, it looks at the **Alerts** index. Here we're looking at time until the past week. We've only just turned this on. Let's look at the past hour to look for network traffic. Again, they're the Alerts in the past hour. We're going to change it from **Alerts**. Uncheck **Alerts**, and check **Traffic**. We can see that there's traffic coming in from a couple of different sources. We've got traffic coming in from our Sophos firewall and our pfSense firewall, but you can now see that there is Modular Sensor traffic coming in. Let's just exclude the other two by doing an include filter, or a filter for, from Modular Sensor traffic. We're only looking at that traffic that's coming in from there. Okay. I can tell you 10.20.1.12 is one of the devices on that 10.20.1 network. That is one of our Active Directory servers that we can see generating traffic.

06:00-06:25
That is the demo for validating that we have turned on network traffic.

The lesson, *Modular Sensor Traffic Monitoring Validation*, is complete. You should now be familiar with network traffic validation. Thank you.